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CAPSTONE PROJECT REPORT — SAP Materials Management

Order-to-Cash (O2C) Process in SAP SD

A comprehensive end-to-end process documentation based on the business scenario of MediSupply Pharma Distributors Pvt. Ltd.

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Topic- Order-to-Cash (O2C) – SAP SD
Module- SAP Materials Management (MM)
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Abstract

This capstone project presents a comprehensive, step-by-step documentation of the Order-to-Cash (O2C) business process implemented within SAP Sales and Distribution (SD), integrated seamlessly with SAP Financial Accounting (FI). The study is based on MediSupply Pharma Distributors Pvt. Ltd., a pharmaceutical distributor handling bulk medicine supply to hospitals and pharmacies. The document traces each stage of the sales lifecycle—from customer master creation through sales order processing, delivery, goods issue, billing, and final payment collection. Special emphasis is placed on batch management, expiry tracking, regulatory compliance, and real-time financial integration. The report highlights how SAP SD automates operations, ensures compliance, and improves efficiency in pharmaceutical supply chains.

1. Introduction

Pharmaceutical distribution requires strict control over product quality, traceability, and compliance. Medicines must be tracked based on batch numbers and expiry dates, making manual processes unreliable. MediSupply Pharma Distributors Pvt. Ltd., headquartered in Hyderabad, operates with over 300 customers including hospitals and retail pharmacies. Prior to SAP implementation, the company relied on spreadsheets and manual systems which resulted in inefficiencies, errors, and compliance risks. SAP SD implementation transformed the organization's operations into a structured and integrated system with real-time data visibility and automated financial postings.

2. Problem Statement

Before SAP implementation, the organization faced numerous operational and financial challenges that affected efficiency and control. There was no centralized customer database, leading to duplicate records and inconsistent customer information. The absence of batch-wise and expiry tracking made it difficult to monitor pharmaceutical inventory, increasing the risk of handling expired products. Billing processes were manual, resulting in frequent errors and incorrect GST calculations. Additionally, poor inventory visibility caused delays in deliveries and inefficient stock management. The lack of integration between sales and finance meant that transactions were not automatically reflected in financial records, requiring manual intervention and increasing the chances of errors. Furthermore, tracking receivables was difficult, leading to delays in collections and poor cash flow management. These challenges collectively resulted in compliance risks, reduced customer satisfaction, and financial losses for the organization.

3. Project Objectives

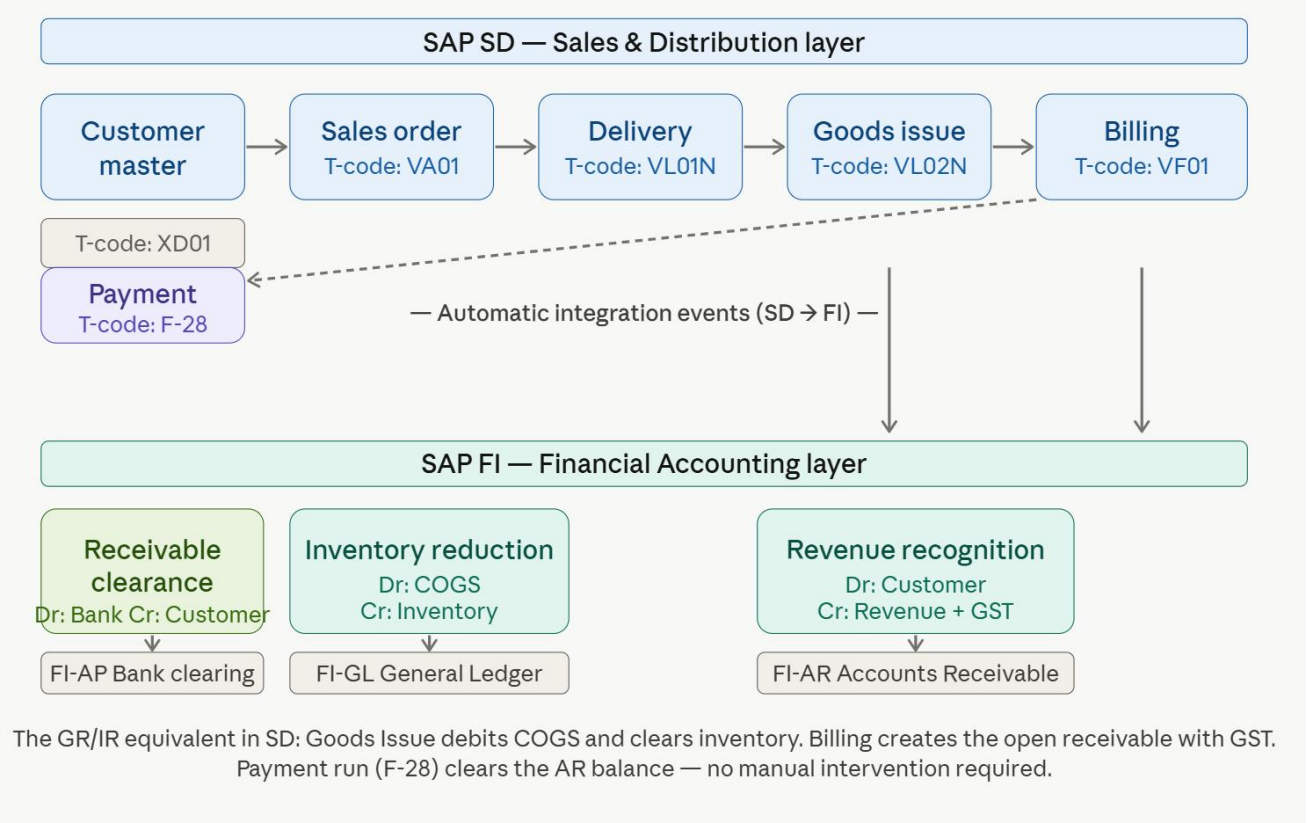
Sl. No.	Objective Statement
1	Establish Centralized Customer Master – Create a unified customer database to ensure consistent, accurate, and duplicate-free customer information across all transactions.
2	Enable Batch-Managed Inventory Tracking – Implement batch tracking to monitor medicines by batch number and expiry for full traceability and compliance.
3	Automate Sales Order Processing – Streamline order creation with automated pricing, tax calculation, and availability checks to reduce errors and improve efficiency.
4	Ensure Expiry-Controlled Delivery – Deliver only valid, non-expired products by integrating expiry checks into the delivery process.
5	Implement GST-Compliant Billing – Generate accurate invoices with automated GST calculation in accordance with tax regulations.
6	Integrate SD with FI for Real-Time Accounting – Enable automatic financial postings from sales transactions to ensure real-time and accurate accounting records.

4. System Architecture – SAP MM & FI Integration

The O2C process operates across SAP SD and SAP FI modules. Each sales transaction automatically triggers corresponding accounting entries, ensuring financial accuracy.
Organizational Units: Company Code: MSP1 Sales Organization: SO-PHARMA Plant: WH-PHARMA Storage Location: MED01 Integration ensures that inventory and financial records are updated simultaneously.

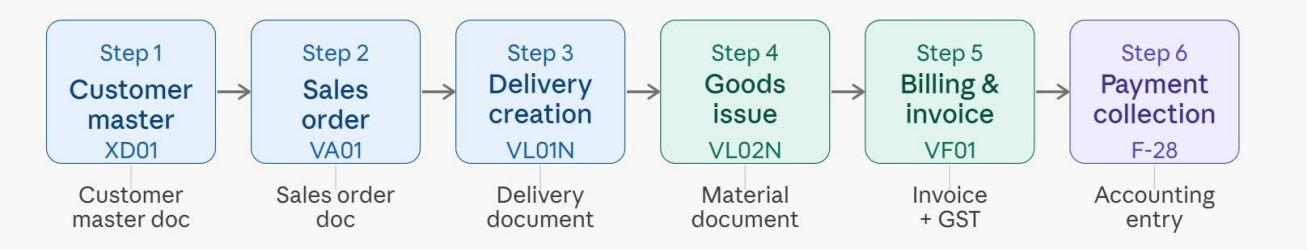
SAP Org Unit	Config. Value	Business Purpose
Company Code	MSP1	Primary legal entity; all financial (FI) postings are recorded here
Sales Organization	SO-PHARMA	Responsible for managing all sales transactions and customer relationships
Distribution Channel	WH (Wholesale)	Defines the channel through which medicines are distributed to pharmacies and hospitals
Division	MED (Medicines)	Categorizes products for pricing, reporting, and sales analysis
Plant	WH01 – Hyderabad	Central warehouse for storage and distribution of pharmaceutical products
Storage Location	MED01 – Finished Goods	Sub-level of plant; stores medicines ready for dispatch
Shipping Point	SP01 – Dispatch Unit	Handles delivery processing, packing, and shipment of goods

4.1 MM–FI Integration Diagram



5. End-to-End Process Flow

The complete Order-to-Cash (O2C) cycle for MediSupply Pharma Distributors Pvt. Ltd. follows a structured six-step sequence as outlined below. Each step is executed through a specific SAP transaction code and generates a corresponding document that seamlessly flows into the subsequent stage of the process.



Step	Process stage	SAP transaction
1.	Customer master creation	XD01
2.	Sales order creation	VA01
3.	Delivery creation	VL01N
4.	Goods issue	VL02N
5.	Billing & invoice	VF01
6.	Payment collection	F-28

6. Detailed Step-by-Step Process Explanation

6.1 Step 1 – Customer Master Data Creation (XD01)

The Order-to-Cash lifecycle begins with the creation of a Customer Master Record, which serves as the central repository of all customer-related information in SAP. For MediSupply Pharma Distributors Pvt. Ltd., a new customer, *CityCare Hospital Pvt. Ltd.*, is onboarded for the supply of pharmaceutical products.

The Customer Master is created using transaction XD01, capturing data across three organisational levels:

Data Level	Key Fields Captured
General Data	Customer name, address, contact details, GSTIN, PAN, bank details
Company Code Data	Reconciliation GL account (FI-AR), payment terms (Net 30), credit limit
Sales Area Data	Sales org (SO-PHARMA), distribution channel (Wholesale), division (Medicines), pricing procedure, shipping conditions

6.1 Step 2 – Sales Order Creation (VA01)

When CityCare Hospital places an order for medicines, a Sales Order (SO) is created using transaction VA01. This is the first formal sales document capturing customer demand.

6.2 Step 3 – Delivery Creation (VL01N)

Based on the sales order, a Delivery Document is created using transaction VL01N.
The warehouse team performs:

- Picking
- Packing
- Batch selection

SAP ensures:

- Only non-expired batches are selected
- Compliance with pharmaceutical regulations

Step 4 – Goods Issue (VL02N)

Once goods are dispatched, Post Goods Issue (PGI) is executed using VL02N.

Action	Detail
Inventory Update	Stock reduced from warehouse
Accounting Entry	Dr: Cost of Goods Sold ₹20,000 Cr: Inventory ₹20,000

This step represents the transfer of ownership of goods to the customer.

6.3 Step 5 – Billing & Invoice Generation (VF01)

The billing document is created using VF01, referencing the delivery document.

Component	Amount
Net Value	₹20,000
GST (12%)	₹2,400
Total Invoice	₹22,400

Step 6 – Consolidated Accounting Flow (PGI → Billing → Payment)

Stage	Entry
Goods Issue	Dr COGS / Cr Inventory
Billing	Dr Customer / Cr Revenue
Payment	Dr Bank / Cr Customer

7. Screenshots & Visual Description

The table below describes the key SAP screen views captured during a live system demonstration, corresponding to each P2P step:

Step	Transaction	Screen Description
Customer Creation	XD01	Customer master screen showing GSTIN, payment terms, and credit limit
Sales Order	VA01	Sales order with material, quantity, pricing details
Delivery	VL01N	Delivery document with batch selection and picking status
Goods Issue	VL02N	PGI confirmation with material and accounting document
Billing	VF01	Invoice screen showing GST calculation
Payment	F-28	Payment clearing screen with bank details

8. Technology Stack

Component	Technology	Detail
ERP Platform	SAP S/4HANA	Latest version
Module	SAP SD	Sales & Distribution
Integration	SAP FI	Financial Accounting
Database	SAP HANA	In-memory processing
Reporting	VA05, VF05	Sales & billing reports
UI	SAP Fiori / GUI	Web + desktop
Output	SmartForms	Invoice generation
Integration	IDocs / EDI	Customer communication

9. Unique Features & Advantages

9.1 Key Features of SAP SD O2C

Feature	Description
Batch Management	Ensures traceability of medicines
Expiry Control	Prevents delivery of expired products
Automatic Billing	GST-compliant invoice generation
Credit Management	Controls customer credit risk
Real-time Integration	SD-FI seamless accounting

9.2 Advantages Delivered to ABC Manufacturing Pvt. Ltd.

Area	Benefit
Accuracy	Eliminates manual errors
Compliance	Ensures regulatory adherence
Speed	Faster order processing
Visibility	Real-time tracking
Finance	Improved cash flow

10. Limitations

Despite its advantages, SAP SD implementation involves high initial costs, complex configuration, and user training requirements. The system's structured workflow may also introduce delays when handling exceptions such as partial deliveries or pricing changes.

11. Future Enhancements

Enhancement	Benefit
AI Demand Forecasting	Better inventory planning
Mobile Sales Apps	Field sales enablement
Barcode/RFID	Faster warehouse operations
E-invoicing Integration	Compliance automation

12. Conclusion

The Order-to-Cash process implemented through SAP SD transforms pharmaceutical distribution into a highly efficient, controlled, and transparent system. Each stage—from customer creation to payment collection—is automated and integrated with financial accounting, ensuring accuracy and compliance.

SAP's real-time integration between SD and FI eliminates manual intervention and provides complete visibility into sales and financial performance, making O2C a critical driver of business success.

References

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[3] KPMG India. ERP Transformation in Pharma Distribution.

[4] SAP Help Portal – SD Module Documentation